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2D Arrays in a Function

ELEC1006: ENGINEERING COMPUTING

2D Array Initialisation

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- Two-dimensional arrays are initialised row-by-row:

```
const int ROWS = 2, COLS = 2;
int exams[ROWS][COLS] = { {84, 78},
                          {92, 97} };
```

- Can omit inner `{ }`, some initial values in a row – array elements without initial values will be set to `0` or `NULL`

2D Array as Argument & Parameter

- Use array name as argument in function call:

```
getExams(exams, 2);
```

- Use empty `[]` for row, size declarator for column in prototype, header:

```
const int COLS = 2;
// Prototype
void getExams(int [][COLS], int);

// Header
void getExams(int exams[][COLS], int rows)
```

Example 1: The `showArray` Function

```
//**********************************************************************
// Function Definition for showArray                                   *
// The first argument is a two-dimensional int array with COLS         *
// columns. The second argument, rows, specifies the number of         *
// rows in the array. The function displays the array's contents.     *
//**********************************************************************

void showArray(int array[][COLS], int rows)
{
    for (int x = 0; x < rows; x++)
    {
        for (int y = 0; y < COLS; y++)
        {
            cout << setw(4) << array[x][y] << " ";
        }
        cout << endl;
    }
}
```

How `showArray` is Called

```
int table1[TBL1_ROWS][COLS] = {{1, 2, 3, 4},
                               {5, 6, 7, 8},
                               {9, 10, 11, 12}};
int table2[TBL2_ROWS][COLS] = {{10, 20, 30, 40},
                               {50, 60, 70, 80},
                               {90, 100, 110, 120},
                               {130, 140, 150, 160}};

cout << "The contents of table1 are:\n";
showArray(table1, TBL1_ROWS);
cout << "The contents of table2 are:\n";
showArray(table2, TBL2_ROWS);
```

Example 1 Output

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Microsoft Visual Studio Debug Console

```
The contents of table1 are:
   1    2    3    4
   5    6    7    8
   9   10   11   12
The contents of table2 are:
  10   20   30   40
  50   60   70   80
  90  100  110  120
 130  140  150  160
```

More info

[1] cplusplus.com: Arrays
https://www.cplusplus.com/doc/tutorial/arrays/

[2] leanrcpp.com: 6.1 – Arrays (Part I)
https://www.learncpp.com/cpp-tutorial/61-arrays-part-i/

[3] learncpp.com: 6.2 – Arrays (Part II)
https://www.learncpp.com/cpp-tutorial/62-arrays-part-ii/

[4] learncpp.com: 6.3 – Arrays and loops
https://www.learncpp.com/cpp-tutorial/63-arrays-and-loops/